Biostatistical Analyses for Injury Phenotype and Risk Assessment

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1 Table One

Table 1: An example of a Table 1

Characteristic	**Flatter**, N = 13[1]	**Hemis**, N = 40[1]	**p-value**	**Overall**[1]
orientation			0.067[2]	

CW	4 (31%)	24 (60%)		28 (53%)
Spine	9 (69%)	16 (40%)		25 (47%)
nom_velocity_m_sec			0.002[3]	
40	13 (100%)	21 (53%)		34 (64%)
70	0 (0%)	19 (48%)		19 (36%)
init_strike_velocity_m_s			0.13[4]	
Mean (SD)	41 (1)	55 (16)		51 (15)
Median (IQR)	41 (40, 42)	42 (40, 71)		42 (40, 70)
Median (Range)	41 (38 - 43)	42 (39 - 75)		42 (38 - 75)
depth_mm			0.73[4]	
Mean (SD)	31 (11)	32 (12)		32 (12)
Median (IQR)	29 (24, 37)	30 (21, 39)		30 (21, 39)
Median (Range)	29 (17 - 53)	30 (18 - 59)		30 (17 - 59)
max_depth_mm			0.94[5]	
Mean (SD)	34 (12)	33 (12)		33 (12)
Median (IQR)	31 (25, 43)	31 (22, 40)		31 (23, 41)
Median (Range)	31 (19 - 57)	31 (19 - 59)		31 (19 - 59)
duration_ms			0.73[4]	
Mean (SD)	1.00 (0.40)	0.96 (0.41)		0.97 (0.40)
Median (IQR)	0.90 (0.73, 1.23)	0.85 (0.62, 1.21)		0.87 (0.62, 1.23)
Median (Range)	0.90 (0.52 - 1.81)	0.85 (0.51 - 2.00)		0.87 (0.51 - 2.00)
transf_energy_j			0.95[5]	
Mean (SD)	124 (41)	135 (84)		130 (68)
Median (IQR)	128 (93, 160)	115 (82, 180)		126 (89, 163)
Median (Range)	128 (67 - 190)	115 (29 - 297)		126 (29 - 297)
Unknown	1	24		25
total_impulse_ns			0.45[5]	
Mean (SD)	3.40 (1.27)	2.96 (1.71)		3.15 (1.53)
Median (IQR)	3.45 (2.43, 4.47)	2.48 (1.49, 4.72)		2.96 (1.80, 4.57)
Median (Range)	3.45 (1.74 - 5.72)	2.48 (0.55 - 5.36)		2.96 (0.55 - 5.72)
Unknown	1	24		25
force_kn			0.30[5]	
Mean (SD)	7.16 (1.48)	7.20 (3.82)		7.19 (3.00)
Median (IQR)	6.82 (5.87, 8.21)	5.22 (4.49, 9.84)		6.19 (5.01, 8.74)
Median (Range)	6.82 (5.37 - 10.26)	5.22 (2.83 - 15.47)		6.19 (2.83 - 15.47)
Unknown	1	24		25
ang_pp_chest_wall_deg			0.095[4]	
Mean (SD)	-13 (9)	-9 (6)		-11 (7)
Median (IQR)	-17 (-20, -2)	-10 (-14, -4)		-11 (-16, -4)
Median (Range)	-17 (-24 - -2)	-10 (-19 - 1)		-11 (-24 - 1)
Unknown	0	12		12
ang_pp_spine_degree			0.077[4]	
Mean (SD)	6.5 (6.8)	10.5 (6.8)		9.2 (7.0)
Median (IQR)	3.0 (1.0, 14.0)	11.0 (5.0, 14.8)		8.0 (3.0, 14.0)
Median (Range)	3.0 (0.0 - 18.0)	11.0 (-2.0 - 25.0)		8.0 (-2.0 - 25.0)
Unknown	0	12		12
aa_traj_chest_wall_deg			0.095[4]	
Mean (SD)	77 (9)	81 (6)		79 (7)
Median (IQR)	73 (70, 88)	80 (77, 86)		79 (74, 86)
Median (Range)	73 (66 - 88)	80 (71 - 91)		79 (66 - 91)
Unknown	0	12		12
aa_traj_spine_degree			0.077[4]	
Mean (SD)	96.5 (6.8)	100.5 (6.8)		99.2 (7.0)
Median (IQR)	93.0 (91.0, 104.0)	101.0 (95.0, 104.8)		98.0 (93.0, 104.0)

Median (Range)	93.0 (90.0 - 108.0)	101.0 (88.0 - 115.0)	98.0 (88.0 - 115.0)
Unknown	0	12	12

[1] n (%)
[2] Pearson's Chi-squared test
[3] Fisher's exact test
[4] Wilcoxon rank sum test
[5] Wilcoxon rank sum exact test

1.1 Cross Refrence

You can cross-reference Table 1 or items in the `bib.bib` file Al-Helali et al. (2021).

You can even reference later figures such as Figure 2 or sections like Section 2 or the page number with 6.

Suppose we wanted to visualize all pairwise t-tests (we will only do first four but you get the idea).

1.2 Pairwise t-tests

1.2.1 between init_strike_velocity_m_s and depth_mm

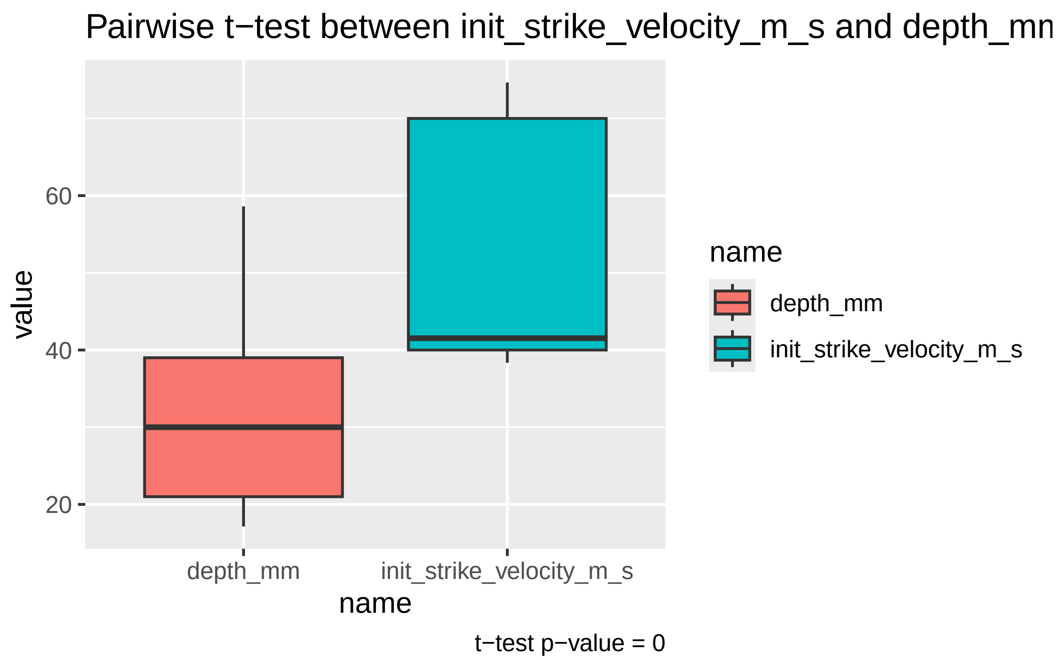

Figure 1: Pairwise t-test between init_strike_velocity_m_s and depth_mm

1.2.2 between init_strike_velocity_m_s and max_depth_mm

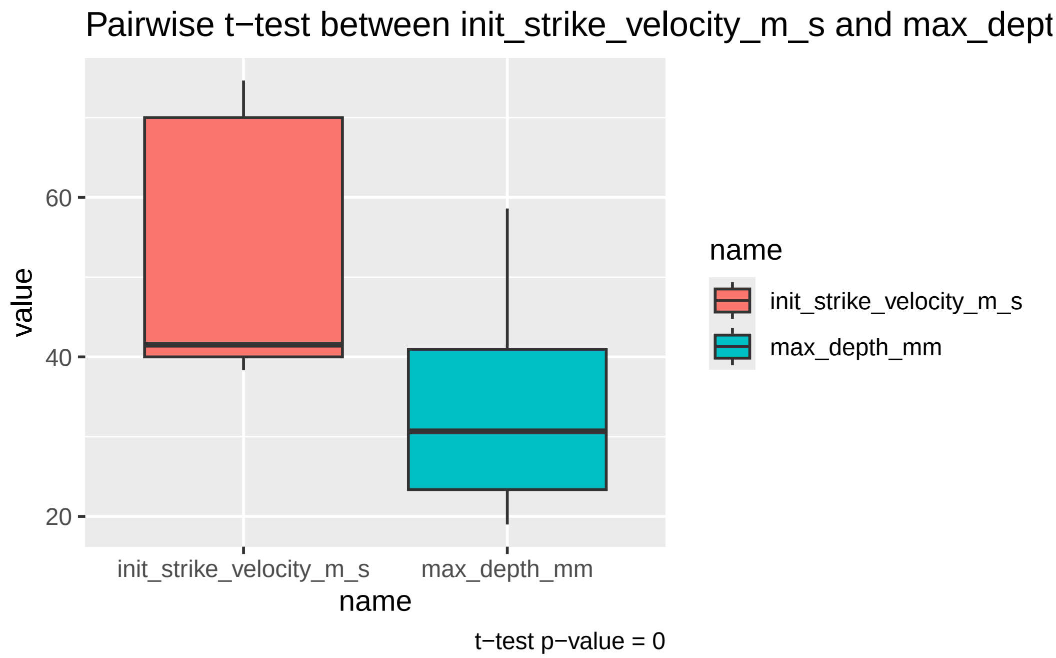

Figure 2: Pairwise t-test between init_strike_velocity_m_s and max_depth_mm

1.2.3 between init__strike__velocity__m__s and duration__ms

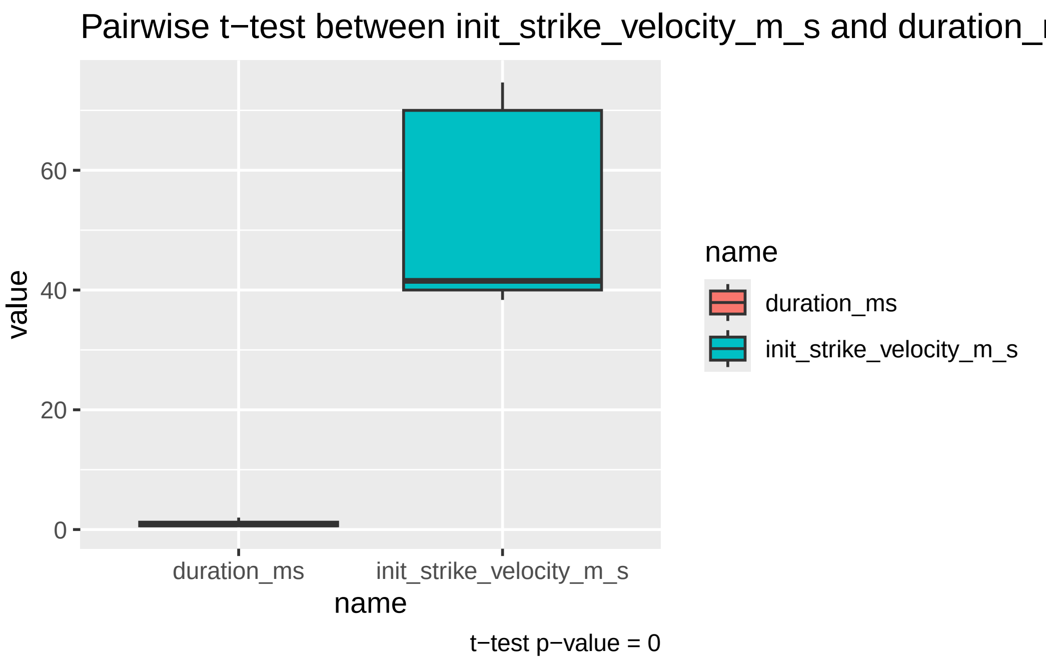

Figure 3: Pairwise t-test between init__strike__velocity__m__s and duration__ms

1.2.4 between init_strike_velocity_m_s and transf_energy_j

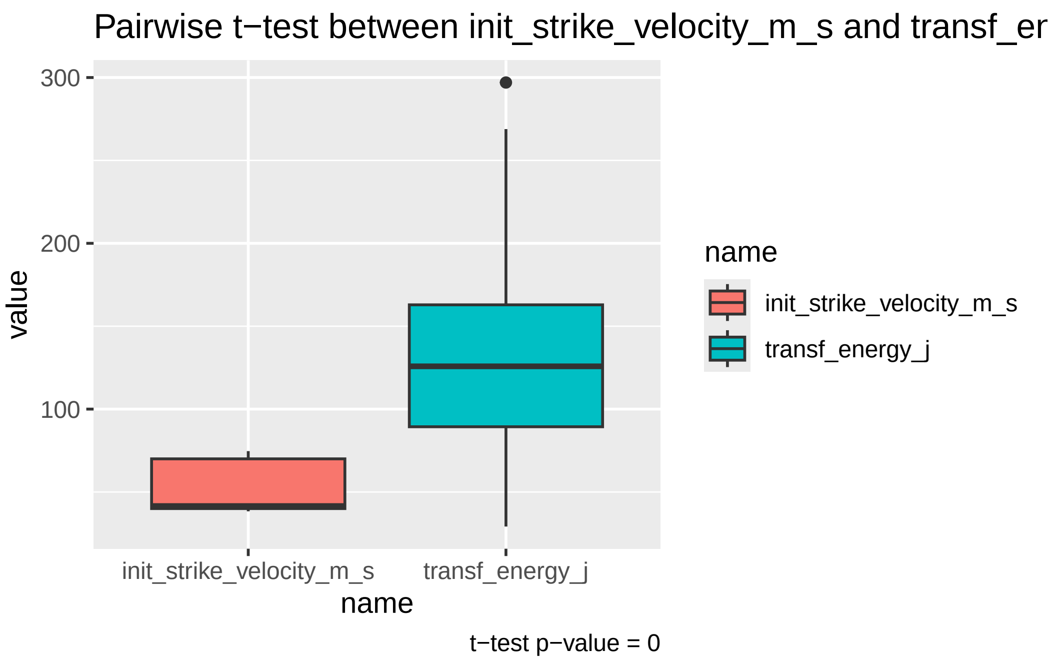

Figure 4: Pairwise t-test between init_strike_velocity_m_s and transf_energy_j

2 Software

This work was completed using R version 4.4.0 (R Core Team 2024) with the following R packages: datasci v. 0.1.0 (Armstrong 2024), grateful v. 0.2.9 (Rodriguez-Sanchez and Jackson 2023), gt v. 0.10.1 (Iannone et al. 2024), gtsummary v. 1.7.2 (Sjoberg et al. 2021), here v. 1.0.1 (Müller 2020), knitr v. 1.47 (Xie 2014, 2015, 2024), rmarkdown v. 2.27 (Xie, Allaire, and Grolemund 2018; Xie, Dervieux, and Riederer 2020; Allaire et al. 2024), tidyverse v. 2.0.0 (Wickham et al. 2019).

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